

Math 421

Wednesday, November 19

Announcements

- Drop-In Hours
 - **Today** 12-1 pm VV 311
 - Thursday 9-11 am VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4
- Homework 9 due Friday

Intermediate Value Theorem

Theorem. Let f be a continuous real-valued function on a closed interval $[a, b]$. If y is a value between $f(a)$ and $f(b)$, then there exists at least one $x \in (a, b)$ such that $f(x) = y$.

Recipe for Solving IVT Problems

- I. Define a function $f(x)$.
- II. Define a number y .
- III. Choose an interval $[a, b]$.
- IV. Establish that f is continuous on $[a, b]$.
- V. Establish that y is between $f(a)$ and $f(b)$.
- VI. Apply the conclusion of the IVT.

Example

1. Show that $x^3 + 2x - 5 = 0$ has a solution.
2. Let $g: [0,1] \rightarrow [0,1]$ be a continuous function. Show that there is some $x_0 \in [0,1]$ such that $g(x_0) = 1 - x_0$.

Activity

1. Show there exists a solution to the equation $3x^5 - 4x^2 = 3$ in the interval $[0,2]$.
2. Prove $xe^x = 2$ for some x in $(0,1)$.

Chapter 3: Continuity

Section 19: Uniform Continuity

Recall – Continuity

- 1 Write down the $\epsilon - \delta$ definition of continuity of a function f at a point x_0 , i.e. that $\lim_{x \rightarrow x_0} f(x) = f(x_0)$.
- 2 What can δ depend on?

Uniform Continuity

Definition. Let f be a real-valued function defined on a set $S \subseteq \mathbb{R}$.
Then f is **uniformly continuous** on S if

Activity – Find δ

1. Show that any constant function is uniformly continuous on $\mathbb{R}$.
2. Let $S = \{x \in \mathbb{R} : -4 \leq x \leq 4\}$, show that $g(x) = x^2$ is uniformly continuous on S .
3. Do you think $g(x) = x^2$ is uniformly continuous on $\mathbb{R}$?

When Does Continuity Imply Uniform Continuity?

Theorem 19.2. If f is a continuous function on $[a, b]$, then f is uniformly continuous on $[a, b]$.

Proof Idea.